🎓 Final Examination – Medicinal Plants

Lecture 5: Leaf Morphology & Microscopy

Time: 2 Hours
Total Marks: 50
Instructions: - Answer all questions. - Choose the best answer. - All questions are based strictly on Lecture 5 content only.

Section A: MCQs (50 Questions)

Q1.

The leaf is best defined as: A. A reproductive organ arising from roots
B. A lateral outgrowth from the stem differing in structure and function
C. An underground modified stem
D. A protective floral organ

Q2.

The three main parts of a typical leaf are: A. Lamina, petiole, margin
B. Apex, base, veins
C. Lamina, petiole, leaf base
D. Petiole, midrib, venation

Q3.

Cotyledonary leaves are mainly concerned with: A. Photosynthesis
B. Protection
C. Storage of food
D. Attraction of insects

Q4.

Prophylls are characterized by being: A. Large and green
B. Reduced and simple in structure
C. Always colored
D. Found only underground

Q5.

Bracts are usually: A. Colorless and membranous
B. Associated with flowers and sometimes colored

C. Found only on roots
D. Always spiny

Q6.

Scale leaves are typically found: A. On aerial stems only
B. On aquatic plants
C. On subterranean stems
D. On floral axes

Q7.

Floral leaves include: A. Foliage leaves only
B. Sepals and reproductive organs
C. Modified stems
D. Cotyledons

Q8.

The most important type of leaf in medicinal plants is: A. Cotyledonary leaf
B. Scale leaf
C. Foliage leaf
D. Floral leaf

Q9.

Modified leaves are primarily adapted for: A. Photosynthesis only
B. Protection or special functions
C. Reproduction
D. Water absorption

Q10.

Which of the following is NOT included in leaf description steps? A. Odour
B. Taste
C. Habitat of plant
D. Texture

Section B: Microscopy & Structure

Q11.

The outermost protective layer of the leaf is: A. Cortex
B. Epidermis
C. Vascular tissue
D. Spongy mesophyll

Q12.

The cuticle is best described as: A. A permeable protein layer
B. A transparent impermeable waxy layer
C. A cellulose tissue
D. A vascular structure

Q13.

Which tissue is mainly concerned with photosynthesis? A. Spongy tissue
B. Palisade tissue
C. Xylem
D. Phloem

Q14.

Spongy mesophyll is characterized by: A. Compact cells without spaces
B. Presence of large intercellular spaces
C. Absence of chloroplasts
D. Thick lignified walls

Q15.

Vascular bundles consist mainly of: A. Cortex and epidermis
B. Xylem and phloem
C. Palisade and spongy tissues
D. Cuticle and epidermis

Section C: Stomata & Epidermis

Q16.

Stomata function mainly in: A. Absorption of minerals
B. Transpiration and gas exchange
C. Mechanical support
D. Storage of food

Q17.

The cells directly surrounding guard cells are called: A. Palisade cells
B. Subsidiary cells
C. Cortex cells
D. Sclerenchyma cells

Q18.

Paracytic stomata are characterized by: A. Subsidiary cells perpendicular to guard cells
B. No subsidiary cells
C. Subsidiary cells parallel to guard cells
D. Unequal subsidiary cells

Q19.

Diacytic stomata have subsidiary cells that are: A. Parallel to guard cells
B. Irregular in shape
C. Perpendicular to guard cells
D. Absent

Q20.

Anisocytic stomata are identified by: A. Two equal subsidiary cells
B. One subsidiary cell smaller than others
C. No subsidiary cells
D. Parallel subsidiary cells

Section D: Epidermal Cell Walls & Cuticle

Q21.

Anticlinal walls may appear all of the following EXCEPT: A. Straight
B. Wavy
C. Beaded
D. Lignified

Q22.

A striated cuticle is characteristically found in: A. Digitalis
B. Saponaria
C. Senna
D. Guava

Q23.

Lower epidermis usually contains: A. More stomata
B. No stomata
C. Vascular bundles
D. Cortex

Q24.

Which cell shape is commonly observed in epidermis? A. Circular
B. Polygonal
C. Cylindrical
D. Cuboidal

Q25.

The primary function of epidermal hairs is: A. Transport of food
B. Protection and secretion
C. Photosynthesis
D. Mechanical strength

🔑 Answer Key (End of File)

1-B | 2-C | 3-C | 4-B | 5-B | 6-C | 7-B | 8-C | 9-B | 10-C
11-B | 12-B | 13-B | 14-B | 15-B
16-B | 17-B | 18-C | 19-C | 20-B
21-D | 22-B | 23-A | 24-B | 25-B

End of Exam

🎓 Final Examination – Medicinal Plants

Lecture 6: Senna & Drugs Acting on GIT

Time: 2 Hours
Total Marks: 50
Instructions: - Answer all questions. - Choose the most appropriate answer. - All questions are strictly based on Lecture 6 content only.

Section A: General Concepts & Terminology

Q1.

Dyspepsia refers to: A. Inflammation of the intestine
B. Defective digestion
C. Excessive bowel movement
D. Chronic constipation

Q2.

Gall stones are formed in the: A. Liver
B. Pancreas
C. Gall bladder
D. Intestine

Q3.

Which term describes a sudden severe pain attack? A. Chronic
B. Acute
C. Colic
D. Dyspepsia

Q4.

Cathartics differ from laxatives in being: A. Slower in action
B. Milder in effect
C. Stronger in purgative action
D. Non-irritant

Q5.

Carminatives are drugs used mainly to: A. Stop diarrhea
B. Increase appetite

C. Expel gases from GIT
D. Treat ulcers

Section B: Senna – Botanical & Morphological Features

Q6.

The biological source of Senna consists of: A. Fresh leaves of Cassia species
B. Dried leaves of Cassia acutifolia
C. Dried roots of Senna alexandrina
D. Dried flowers of Cassia senna

Q7.

Senna belongs to which family? A. Solanaceae
B. Lamiaceae
C. Fabaceae (Leguminosae)
D. Myrtaceae

Q8.

The leaves of Senna are described as: A. Simple leaves
B. Compound leaves with leaflets
C. Scale leaves
D. Modified spines

Q9.

The phyllotaxis of Senna leaves is: A. Opposite
B. Whorled
C. Alternate
D. Radical

Q10.

The shape of Senna leaflets is usually: A. Linear
B. Cordate
C. Lanceolate to ovate-lanceolate
D. Orbicular

Section C: Active Constituents

Q11.

The major active constituents of Senna are: A. Alkaloids
B. Cardiac glycosides
C. Anthraquinone glycosides
D. Flavonoids only

Q12.

Sennosides A and B are chemically classified as: A. Free anthraquinones
B. Dianthrone glycosides
C. Flavone glycosides
D. Triterpenes

Q13.

The active metabolite responsible for Senna action is: A. Aloe-emodin
B. Rhein aglycone
C. Chrysophanol
D. Emodin

Q14.

Sennosides are considered prodrugs because they: A. Act directly in the stomach
B. Are inactive until metabolized by intestinal bacteria
C. Are rapidly absorbed in the upper GIT
D. Are enzymatically degraded in the liver

Q15.

Senna glycosides are mainly absorbed in the: A. Stomach
B. Small intestine
C. Large intestine
D. Colon after bacterial action

Section D: Mechanism of Action

Q16.

Senna produces laxative action mainly by: A. Decreasing water secretion in colon
B. Inhibiting peristalsis
C. Stimulating peristalsis of the colon
D. Relaxing intestinal smooth muscles

Q17.

Senna increases bowel movement by enhancing: A. Sodium absorption
B. Chloride secretion into colon
C. Potassium retention
D. Calcium reabsorption

Q18.

The increased water content in stool is due to: A. Active transport
B. Osmotic effect
C. Diffusion only
D. Enzymatic hydrolysis

Q19.

Which condition is MOST appropriate for Senna use? A. Chronic constipation
B. Intestinal obstruction
C. Occasional acute constipation
D. Children below 6 years

Q20.

Long-term use of Senna may lead to: A. Hyperkalemia
B. Hypokalemia
C. Hypercalcemia
D. Dehydration only

Section E: Uses, Contraindications & Side Effects

Q21.

Senna is contraindicated in: A. Hemorrhoids
B. Post-operative bowel evacuation
C. Intestinal obstruction
D. Radiological examination of colon

Q22.

One of the common side effects of Senna is: A. Sedation
B. Griping abdominal pain
C. Respiratory depression
D. Hypotension

Q23.

Loss of potassium due to Senna may affect mainly: A. Liver function
B. Kidney filtration
C. Cardiac and muscular activity
D. Brain neurotransmission

Q24.

Senna should NOT be used chronically because it: A. Causes tolerance
B. Leads to electrolyte imbalance
C. Is poorly absorbed
D. Loses efficacy rapidly

Q25.

Which statement about Senna is CORRECT? A. It acts in the stomach B. It is safe for prolonged use C. It is activated by colonic bacteria D. It decreases intestinal secretion

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🔑 Answer Key (End of File)

1-B | 2-C | 3-B | 4-C | 5-C
6-B | 7-C | 8-B | 9-C | 10-C
11-C | 12-B | 13-B | 14-B | 15-D
16-C | 17-B | 18-B | 19-C | 20-B
21-C | 22-B | 23-C | 24-B | 25-C

End of Exam

Final Examination – Medicinal Plants

Lecture 7: ANS – Belladonna – Datura – Pilocarpus

Time: 2 Hours
Total Marks: 50
Instructions: - Answer all questions. - Choose the best answer. - All questions are strictly based on Lecture 7 content only.

Section A: Autonomic Nervous System (ANS)

Q1.

The autonomic nervous system is mainly concerned with: A. Voluntary skeletal muscle movements
B. Conscious sensory perception
C. Involuntary activities of organs and glands
D. Higher mental functions

Q2.

The parasympathetic system is predominantly active during: A. Fear and stress
B. Exercise
C. Rest and digestion
D. Emergency situations

Q3.

Which neurotransmitter is associated with parasympathetic activity? A. Adrenaline
B. Noradrenaline
C. Acetylcholine
D. Dopamine

Q4.

Sympathetic stimulation causes all EXCEPT: A. Pupil dilation
B. Increased heart rate
C. Increased gastrointestinal motility
D. Bronchodilation

Q5.

Anticholinergic drugs mainly block the action of: A. Adrenaline
B. Acetylcholine

C. Noradrenaline
D. Serotonin

Section B: Belladonna & Datura – General Concepts

Q6.

Atropa belladonna and Datura stramonium belong to the family: A. Fabaceae
B. Myrtaceae
C. Solanaceae
D. Apiaceae

Q7.

The main pharmacological effect of both Belladonna and Datura is: A. Cholinergic stimulation
B. Sympathomimetic action
C. Anticholinergic action
D. Sedative action

Q8.

The popular name “Deadly nightshade” refers to: A. Datura stramonium
B. Atropa belladonna
C. Pilocarpus jaborandi
D. Hyoscyamus

Q9.

Datura stramonium is commonly associated with: A. Pleasant odor and taste
B. CNS depression only
C. Hallucinations and confusion
D. Cardiac glycosides

Q10.

Which plant is commonly known as Devil’s apple? A. Belladonna
B. Datura stramonium
C. Pilocarpus
D. Hyoscyamus

Section C: Morphology & Organoleptic Characters

Q11.

Belladonna leaves are best described as: A. Narrow lanceolate
B. Ovate to broadly ovate
C. Linear
D. Cordate

Q12.

The odor of Datura leaves is usually: A. Pleasant aromatic
B. Odorless
C. Slight but unpleasant
D. Strongly fruity

Q13.

The taste of Belladonna leaves is: A. Sweet
B. Tasteless
C. Bitter
D. Salty

Q14.

A major distinguishing microscopic feature of Datura is: A. Presence of idioblasts only
B. Crystal layer of calcium oxalate
C. Absence of stomata
D. Oil glands

Q15.

Belladonna is microscopically characterized by: A. Beaded anticlinal walls
B. Idioblast containing crystals
C. Multicellular glandular hairs only
D. Absence of trichomes

Section D: Pilocarpus (Jaborandi)

Q16.

Pilocarpus primarily produces which pharmacological effect? A. Anticholinergic
B. Cholinergic (parasympathomimetic)
C. Sympathomimetic
D. CNS stimulant

Q17.

Pilocarpine is mainly used clinically to treat: A. Constipation
B. Hypertension
C. Glaucoma
D. Insomnia

Q18.

Pilocarpus differs from Belladonna in that it: A. Causes dry mouth
B. Increases salivary secretion
C. Produces hallucinations
D. Causes urinary retention

Q19.

Which effect is expected with Pilocarpus administration? A. Mydriasis
B. Decreased sweating
C. Increased lacrimation
D. Bronchodilation

Q20.

Pilocarpus belongs to the family: A. Solanaceae
B. Rutaceae
C. Fabaceae
D. Myrtaceae

Section E: Toxicity & Clinical Correlation

Q21.

A classic sign of anticholinergic toxicity is: A. Diarrhea
B. Excess salivation
C. Dry mouth and blurred vision
D. Bradycardia

Q22.

The phrase “dry as a bone” refers to: A. Severe dehydration
B. Reduced sweating and secretions
C. Bone pain
D. Muscle rigidity

Q23.

Anticholinergic overdose may lead to: A. Hypotension and diarrhea
B. Confusion and hallucinations
C. Bronchoconstriction
D. Increased intestinal motility

Q24.

Which condition is a contraindication for anticholinergic drugs? A. Motion sickness
B. Parkinsonism
C. Glaucoma
D. Bradycardia

Q25.

Which statement is CORRECT? A. Belladonna stimulates parasympathetic activity
B. Datura produces cholinergic effects
C. Pilocarpus enhances acetylcholine action
D. All act via the same mechanism

🔑 Answer Key (End of File)

1-C | 2-C | 3-C | 4-C | 5-B
6-C | 7-C | 8-B | 9-C | 10-B
11-B | 12-C | 13-C | 14-B | 15-B
16-B | 17-C | 18-B | 19-C | 20-B
21-C | 22-B | 23-B | 24-C | 25-C

End of Exam

🎓 Final Examination – Medicinal Plants

Lecture 8: Hyoscyamus – Tropane Alkaloids – Tea Leaves

Time: 2 Hours
Total Marks: 50
Instructions: - Answer all questions. - Choose the most appropriate answer. - All questions are strictly based on Lecture 8 content only.

Section A: Hyoscyamus – General Description

Q1.

Hyoscyamus muticus belongs to the family: A. Fabaceae
B. Solanaceae
C. Rutaceae
D. Myrtaceae

Q2.

Hyoscyamus leaves are macroscopically described as: A. Narrow and linear
B. Small and leathery
C. Large, broad and succulent
D. Scale-like

Q3.

The succulent nature of Hyoscyamus leaves is mainly due to: A. High oil content
B. Desert adaptation for water storage
C. Excess cellulose
D. Thick cuticle only

Q4.

Hyoscyamus is commonly known in Arabic as: A. الحرنكش
B. السكران المصري
C. العرقسوس
D. الحناء

Q5.

The main therapeutic interest of Hyoscyamus is related to its action on: A. Cardiovascular system
B. Respiratory system

C. Central nervous system
D. Renal system

Section B: Active Constituents & Chemical Tests

Q6.

The principal active constituents of Hyoscyamus are: A. Cardiac glycosides
B. Tropane alkaloids
C. Anthraquinones
D. Flavonoids

Q7.

Which alkaloid is naturally present in Hyoscyamus? A. Atropine
B. Hyoscine (Scopolamine)
C. Morphine
D. Pilocarpine

Q8.

Atropine may be formed from Hyoscyamus alkaloids by: A. Enzymatic oxidation
B. Exposure to harsh conditions causing isomerization
C. Bacterial metabolism
D. Acid hydrolysis

Q9.

The general test for alkaloids gives a creamy white precipitate with: A. Dragendorff's reagent
B. Vitali's test
C. Mayer's reagent
D. Wagner's reagent

Q10.

Vitali–Morin reaction is specific for: A. Indole alkaloids
B. Purine alkaloids
C. Tropane alkaloids
D. Cardiac glycosides

Section C: Pharmacological Actions

Q11.

Hyoscyamus alkaloids primarily exhibit: A. Cholinergic action
B. Anticholinergic action
C. Adrenergic blocking action
D. Cardiotonic effect

Q12.

Which CNS effect is associated with Hyoscine? A. CNS stimulation
B. Sedative and antiemetic effect
C. Antidepressant action
D. Anticonvulsant action

Q13.

Hyoscyamus reduces gastrointestinal motility by: A. Increasing acetylcholine release
B. Blocking muscarinic receptors
C. Stimulating sympathetic nerves
D. Increasing enzyme secretion

Q14.

Which of the following is a therapeutic use of Hyoscyamus? A. Chronic constipation
B. Motion sickness
C. Hypertension
D. Asthma

Q15.

Hyoscine is preferred over atropine mainly because it: A. Is more potent on the heart
B. Has stronger stimulant effect
C. Produces more CNS sedation
D. Has no side effects

Section D: Adverse Effects & Toxicity

Q16.

A common adverse effect of Hyoscyamus is: A. Excess salivation
B. Diarrhea
C. Dry mouth
D. Bradycardia

Q17.

The term “anticholinergic crisis” refers to: A. Enhanced acetylcholine activity
B. Severe blockage of muscarinic receptors
C. Reduced sympathetic tone
D. Increased parasympathetic discharge

Q18.

Which symptom is included in the mnemonic HORSE MAD? A. Hypoglycemia
B. Miosis
C. Dry skin and hyperthermia
D. Diarrhea

Q19.

Hyoscyamus is contraindicated in patients with: A. Peptic ulcer
B. Glaucoma
C. Motion sickness
D. Parkinsonism

Q20.

Orthostatic hypotension means: A. High blood pressure while standing
B. Sudden drop in blood pressure on standing
C. Increased heart rate during exercise
D. Chronic hypotension

Section E: Tea Leaves (Camellia sinensis)

Q21.

Tea leaves are obtained from: A. Mature lower leaves only
B. Young terminal leaves and buds
C. Roots of the plant
D. Stems and bark

Q22.

The best quality tea is collected when the leaves are: A. Fully mature
B. Old and fibrous
C. Young and tender
D. Dried on the plant

Q23.

Tea belongs to the family: A. Solanaceae
B. Rutaceae
C. Theaceae
D. Myrtaceae

Q24.

Which compound contributes mainly to the stimulant effect of tea? A. Tannins
B. Caffeine
C. Flavonoids
D. Saponins

Q25.

Which statement about tea leaves is CORRECT? A. Older leaves give better quality tea
B. Tea is devoid of alkaloids
C. Quality decreases with leaf maturity
D. Tea has no CNS effect

🔑 Answer Key (End of File)

1-B | 2-C | 3-B | 4-B | 5-C
6-B | 7-B | 8-B | 9-C | 10-C
11-B | 12-B | 13-B | 14-B | 15-C
16-C | 17-B | 18-C | 19-B | 20-B
21-B | 22-C | 23-C | 24-B | 25-C

End of Exam

🎓 Final Examination – Medicinal Plants

Lecture 9: Uva Ursi & Plants Acting on Urinary Tract

Time: 2 Hours
Total Marks: 50
Instructions: - Answer all questions. - Choose the most appropriate answer. - All questions are strictly based on Lecture 9 content only.

Section A: General Information

Q1.

Uva Ursi is commonly known as: A. Bearberry
B. Foxglove
C. Deadly nightshade
D. Henbane

Q2.

Uva Ursi belongs to the family: A. Solanaceae
B. Myrtaceae
C. Ericaceae
D. Fabaceae

Q3.

The official drug of Uva Ursi consists mainly of: A. Roots
B. Fruits
C. Leaves
D. Bark

Q4.

The minimum acceptable percentage of active constituents in Uva Ursi is: A. 2%
B. 5%
C. 8%
D. 10%

Q5.

The major therapeutic application of Uva Ursi is in treating: A. Respiratory infections
B. Urinary tract infections

C. Cardiac disorders
D. CNS disorders

Section B: Active Constituents

Q6.

The main active constituent of Uva Ursi is: A. Caffeine
B. Arbutin
C. Sennosides
D. Digitoxin

Q7.

Arbutin is chemically classified as: A. Alkaloid
B. Phenolic glycoside
C. Cardiac glycoside
D. Flavonoid

Q8.

Hydroquinone is released from arbutin by: A. Acid hydrolysis in stomach
B. Enzymatic hydrolysis in intestine
C. Bacterial metabolism in colon
D. Liver metabolism

Q9.

Which compound is responsible for antibacterial action in Uva Ursi? A. Gallic acid
B. Hydroquinone
C. Triterpenes
D. Tannins only

Q10.

Which additional constituent contributes to astringent action? A. Flavonoids
B. Triterpenes
C. Tannins
D. Alkaloids

Section C: Chemical Tests

Q11.

Sublimation test for Uva Ursi gives crystals that are: A. Colorless needles
B. Brown rods
C. White plates
D. Yellow prisms

Q12.

Ferric chloride test for Uva Ursi gives a color: A. Green
B. Blue-black
C. Bluish violet
D. Red

Q13.

Arbutin methyl differs from arbutin by the presence of: A. Additional glucose unit
B. Methoxy group
C. Sulfate group
D. Acetyl group

Q14.

The phenolic nature of Uva Ursi constituents is confirmed by: A. Mayer’s test
B. Vitali test
C. FeCl3 test
D. Dragendorff’s test

Q15.

Which test confirms the presence of hydroquinone derivatives? A. Sublimation test
B. Molisch test
C. Biuret test
D. Ninhydrin test

Section D: Uses & Pharmacological Actions

Q16.

Uva Ursi acts in UTI mainly by: A. Increasing urine acidity
B. Antibacterial action in urinary tract
C. Diuretic effect only
D. CNS depression

Q17.

The astringent effect of Uva Ursi helps by: A. Increasing urine flow
B. Reducing mucosal inflammation
C. Stimulating bladder muscles
D. Relaxing smooth muscles

Q18.

Which condition may discolor urine during Uva Ursi use? A. Renal failure
B. Oxidation of hydroquinone
C. Excess tannins
D. Dehydration

Q19.

Uva Ursi should be avoided for long-term use because it: A. Is ineffective
B. Causes liver toxicity
C. Contains high tannin content
D. Produces tolerance

Q20.

Which patient should avoid Uva Ursi? A. Acute cystitis patient
B. Patient with gastric irritation
C. Mild UTI patient
D. Adult female patient

Section E: Adverse Effects & Precautions

Q21.

A common adverse effect due to tannins is: A. Diarrhea
B. Constipation
C. Excess salivation
D. Hypotension

Q22.

Long-term hydroquinone exposure may lead to: A. Renal toxicity
B. Hepatotoxicity
C. Neurotoxicity
D. Cardiotoxicity

Q23.

The antibacterial action of Uva Ursi is optimal when urine is: A. Acidic
B. Neutral
C. Alkaline
D. Concentrated

Q24.

Which statement about Uva Ursi is CORRECT? A. Safe for prolonged use
B. Used mainly as laxative
C. Acts locally in urinary tract
D. Free of phenolic compounds

Q25.

The discoloration of urine during therapy is usually: A. Dangerous
B. Due to infection
C. Harmless and reversible
D. Permanent

🔍 Answer Key (End of File)

1-A | 2-C | 3-C | 4-C | 5-B
6-B | 7-B | 8-B | 9-B | 10-C
11-B | 12-C | 13-B | 14-C | 15-A
16-B | 17-B | 18-B | 19-C | 20-B
21-B | 22-B | 23-C | 24-C | 25-C

End of Exam

🎓 Final Examination – Medicinal Plants

Lecture 10: Plants Acting on Respiratory Tract

Time: 2 Hours
Total Marks: 50
Instructions: - Answer all questions. - Choose the most appropriate answer. - All questions are strictly based on Lecture 10 content only.

Section A: General Concepts

Q1.

Respiratory tract medicinal plants are mainly used to treat: A. Cardiac disorders
B. Gastrointestinal disorders
C. Cough and respiratory complaints
D. Urinary tract infections

Q2.

Cough associated with sputum is described as: A. Dry cough
B. Productive cough
C. Allergic cough
D. Reflex cough

Q3.

An expectorant primarily helps by: A. Suppressing cough reflex
B. Increasing sputum viscosity
C. Facilitating expulsion of mucus
D. Causing bronchodilation

Q4.

Demulcents act mainly by: A. Bronchodilation
B. Reducing irritation by coating mucosa
C. Increasing secretion
D. Antibacterial action

Q5.

Which term describes inflammation of the gastrointestinal tract with diarrhea? A. Dysentery
B. Gastroenteritis

C. Dyspepsia
D. Colitis

Section B: Guava Leaves (Psidium guajava)

Q6.

Guava leaves belong to the family: A. Solanaceae
B. Fabaceae
C. Myrtaceae
D. Rutaceae

Q7.

Guava leaves are characterized by high content of: A. Alkaloids
B. Cardiac glycosides
C. Tannins
D. Saponins

Q8.

The astringent property of guava leaves is mainly useful in treating: A. Constipation
B. Diarrhea
C. Hypertension
D. Asthma

Q9.

Guava leaves are commonly used as gargle due to their: A. Antiviral effect
B. Antibacterial and deodorant action
C. Sedative effect
D. Diuretic action

Q10.

The taste of guava leaves is best described as: A. Sweet
B. Bitter
C. Astringent
D. Sour

Section C: Eucalyptus

Q11.

Eucalyptus leaves are mainly valued for their content of: A. Alkaloids
B. Volatile oil (Eucalyptus oil)
C. Cardiac glycosides
D. Anthraquinones

Q12.

Eucalyptus oil produces a sensation of coolness mainly by: A. Reducing blood flow
B. Acting on cold receptors
C. Increasing sweating
D. Depressing CNS

Q13.

A major therapeutic use of eucalyptus oil is: A. Laxative
B. Expectorant and decongestant
C. Cardiotonic
D. Sedative

Q14.

The characteristic odor of eucalyptus is due to: A. Tannins
B. Flavonoids
C. Volatile oil
D. Glycosides

Q15.

Which animal is uniquely adapted to feed on eucalyptus leaves? A. Camel
B. Goat
C. Koala
D. Deer

Section D: Ivy (Hedera helix)

Q16.

Ivy belongs to the family: A. Myrtaceae
B. Solanaceae
C. Araliaceae
D. Fabaceae

Q17.

The major active constituents of ivy are: A. Alkaloids
B. Saponins
C. Cardiac glycosides
D. Flavonoids only

Q18.

Ivy is mainly used in respiratory disorders as: A. Antitussive
B. Expectorant and mucolytic
C. Bronchoconstrictor
D. Sedative

Q19.

Saponins exert expectorant action mainly by: A. Direct bronchodilation
B. Reflex irritation of gastric mucosa
C. CNS depression
D. Antibacterial action

Q20.

Which statement about ivy is CORRECT? A. Used only for dry cough
B. Safe without side effects
C. Enhances mucus secretion and expulsion
D. Causes constipation

Section E: Clinical Correlation & Safety

Q21.

Which plant is most suitable for productive cough? A. Guava
B. Eucalyptus
C. Ivy
D. Senna

Q22.

Overuse of eucalyptus oil may cause: A. Sedation
B. Respiratory irritation
C. Hypotension
D. Bradycardia

Q23.

Which preparation is MOST appropriate for relieving nasal congestion? A. Guava leaf decoction
B. Ivy syrup
C. Eucalyptus inhalation
D. Senna infusion

Q24.

Demulcent action is especially beneficial in: A. Dry cough
B. Productive cough
C. Cardiac asthma
D. Renal cough

Q25.

Which statement about respiratory plants is CORRECT? A. All suppress cough reflex
B. All act as antibiotics
C. Selection depends on type of cough
D. They are interchangeable

🔍 Answer Key (End of File)

1-C | 2-B | 3-C | 4-B | 5-B
6-C | 7-C | 8-B | 9-B | 10-C
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21-C | 22-B | 23-C | 24-A | 25-C

End of Exam

🎓 Final Examination – Medicinal Plants

Lecture 11: Cardiac Glycosides – Digitalis

Time: 2 Hours
Total Marks: 50
Instructions: - Answer all questions. - Choose the most appropriate answer. - All questions are strictly based on Lecture 11 content only.

Section A: General Concepts

Q1.

Digitalis is mainly used in the treatment of: A. Respiratory disorders
B. Cardiac insufficiency
C. CNS disorders
D. Urinary tract infections

Q2.

Digitalis plants belong to the family: A. Solanaceae
B. Fabaceae
C. Scrophulariaceae
D. Myrtaceae

Q3.

The official Digitalis species mentioned in pharmacopoeia is: A. Digitalis lanata
B. Digitalis purpurea
C. Digitalis lutea
D. Digitalis ferruginea

Q4.

The best time for collection of Digitalis leaves is: A. After flowering
B. Before flowering or at the beginning of flowering
C. During fruiting
D. Any time of the year

Q5.

Rapid drying of Digitalis leaves is necessary to: A. Improve color
B. Prevent microbial growth

C. Prevent enzymatic degradation of glycosides
D. Increase moisture content

Section B: Morphology & Microscopy

Q6.

Digitalis purpurea leaves are typically: A. Lanceolate and woolly
B. Ovate with smooth texture
C. Linear and glabrous
D. Scale-like

Q7.

Digitalis lanata leaves are characterized by being: A. Smooth and hairless
B. Woolly and lanceolate
C. Ovate and glabrous
D. Cordate

Q8.

Both Digitalis species possess which type of stomata? A. Paracytic
B. Diacytic
C. Anomocytic
D. Anisocytic

Q9.

The anticlinal walls in Digitalis lanata are described as: A. Straight
B. Smooth
C. Beaded
D. Thickened

Q10.

Which trichomes are found in Digitalis leaves? A. Non-glandular only
B. Glandular only
C. Both glandular and non-glandular
D. No trichomes

Section C: Chemical Structure of Cardiac Glycosides

Q11.

Cardiac glycosides consist of: A. Sugar part only
B. Steroidal nucleus only
C. Glycone and aglycone parts
D. Alkaloid structure

Q12.

The aglycone part of Digitalis glycosides is: A. Flavonoid nucleus
B. Alkaloid ring
C. Steroidal nucleus with lactone ring
D. Triterpene structure

Q13.

Digitoxose is classified as: A. Pentose sugar
B. Deoxy sugar
C. Amino sugar
D. Disaccharide

Q14.

Cardenolides are characterized by the presence of: A. Six-membered lactone ring
B. Five-membered lactone ring
C. Aromatic ring
D. Indole nucleus

Q15.

The sugar part of Digitalis glycosides is responsible mainly for: A. Pharmacological activity
B. Toxicity
C. Solubility and pharmacokinetics
D. Enzyme inhibition

Section D: Enzymatic & Acid Hydrolysis

Q16.

Enzymatic hydrolysis of purpurea glycosides results in removal of: A. All sugar units
B. Terminal glucose only
C. Lactone ring
D. Steroidal nucleus

Q17.

Acid hydrolysis of Digitalis glycosides leads to: A. Partial hydrolysis
B. Removal of glucose only
C. Complete removal of all sugars
D. No effect

Q18.

Purpurea A glycoside ultimately yields which active compound? A. Digoxin
B. Digitoxin
C. Gitoxin
D. Lanatoside C

Q19.

Lanatosides differ from purpurea glycosides by the presence of: A. Hydroxyl group
B. Methyl group
C. Acetyl group
D. Sulfate group

Q20.

Gitoxin is derived from: A. Purpurea B glycoside
B. Lanata A glycoside
C. Purpurea A glycoside
D. Lanata C glycoside

Section E: Pharmacological Action & Safety

Q21.

The main therapeutic effect of Digitalis is: A. Increase heart rate
B. Decrease myocardial contractility
C. Increase force of cardiac contraction
D. Vasodilation

Q22.

Digitalis toxicity is most likely associated with: A. Hypocalcemia
B. Hyperkalemia
C. Narrow therapeutic index
D. High solubility

Q23.

Which symptom is commonly associated with Digitalis toxicity? A. Diarrhea only
B. Visual disturbances
C. Bronchospasm
D. Polyuria

Q24.

Digitalis should be used cautiously because it: A. Acts rapidly
B. Accumulates in the body
C. Is rapidly excreted
D. Has low potency

Q25.

Which statement about Digitalis is CORRECT? A. Safe for long-term unsupervised use
B. Used mainly for hypertension
C. Requires dose monitoring
D. Acts as diuretic only

🔑 Answer Key (End of File)

1-B | 2-C | 3-B | 4-B | 5-C
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End of Exam

🎓 Final Examination – Medicinal Plants

Lecture 12: Plant Taxonomy

Time: 2 Hours
Total Marks: 50
Instructions: - Answer all questions. - Choose the most appropriate answer. - All questions are strictly based on Lecture 12 content only.

Section A: General Classification Concepts

Q1.

Living organisms are classified into how many kingdoms? A. Three
B. Four
C. Five
D. Six

Q2.

Which of the following kingdoms was previously considered part of the plant kingdom? A. Animalia
B. Fungi
C. Protista
D. Monera

Q3.

Fungi differ from plants mainly because they: A. Lack a cell wall
B. Contain cellulose
C. Do not contain chlorophyll
D. Are multicellular only

Q4.

The cell wall of fungi is mainly composed of: A. Cellulose
B. Lignin
C. Chitin
D. Pectin

Q5.

Which feature is ESSENTIAL for an organism to be classified as a plant? A. Motility
B. Ability to synthesize food

C. Presence of nucleus only
D. Ability to ingest food

Section B: Definition & Importance of Plant Taxonomy

Q6.

Plant taxonomy is the science concerned with: A. Plant physiology only
B. Plant chemistry only
C. Identification, nomenclature, and classification
D. Medicinal uses of plants

Q7.

The term “taxonomy” is derived from Greek words meaning: A. Name and form
B. Order and method
C. Plant and life
D. Growth and structure

Q8.

The scientist who established the binomial system of nomenclature is: A. Darwin
B. Mendel
C. Carolus Linnaeus
D. Aristotle

Q9.

The binomial system consists of: A. Genus and species
B. Family and genus
C. Order and family
D. Species and variety

Q10.

Scientific names of plants are usually written in: A. Arabic language
B. English language
C. Latin language
D. French language

Section C: Major Plant Groups

Q11.

Cryptogams are characterized by: A. Presence of flowers
B. Absence of flowers and seeds
C. Presence of fruits
D. Covered seeds

Q12.

Which group of plants reproduces by spores only? A. Angiosperms
B. Gymnosperms
C. Cryptogams
D. Dicotyledons

Q13.

Thallophyta are characterized by: A. Differentiated roots, stems, and leaves
B. Vascular tissues
C. Undifferentiated plant body
D. Presence of flowers

Q14.

Bryophytes are best described as: A. Vascular plants with seeds
B. Non-vascular plants with root-like structures
C. Flowering plants
D. Plants with covered seeds

Q15.

Pteridophytes differ from bryophytes by having: A. Flowers
B. Seeds
C. Vascular tissues
D. Fruits

Section D: Seed Plants

Q16.

Gymnosperms are characterized by: A. Covered seeds
B. Fruits formation
C. Naked seeds
D. Absence of vascular tissues

Q17.

Angiosperms are plants that: A. Reproduce by spores only
B. Have naked seeds
C. Bear flowers and fruits
D. Lack vascular tissues

Q18.

Angiosperms are divided into monocotyledons and dicotyledons based on: A. Type of root
B. Leaf venation
C. Number of cotyledons
D. Type of flower

Q19.

Monocotyledon plants usually have: A. Reticulate venation
B. Two cotyledons
C. Parallel venation
D. Secondary growth

Q20.

Dicotyledon plants are characterized by: A. One cotyledon
B. Parallel venation
C. Reticulate venation
D. Absence of vascular tissue

Section E: Taxonomic Hierarchy & Applications

Q21.

The correct order of taxonomic hierarchy from highest to lowest is: A. Species → Genus → Family → Order → Class → Phylum → Kingdom
B. Kingdom → Phylum → Class → Order → Family → Genus → Species
C. Kingdom → Class → Phylum → Family → Order → Species → Genus
D. Phylum → Kingdom → Class → Order → Family → Genus → Species

Q22.

Plant taxonomy is important in medicinal plants because it helps in: A. Increasing plant yield
B. Avoiding adulteration and misidentification
C. Improving taste of drugs
D. Increasing toxicity

Q23.

Belladonna, Datura, and Hyoscyamus belong to the same family because they share: A. Same habitat
B. Similar flower color
C. Similar active constituents
D. Same geographical origin

Q24.

Which of the following is the LOWEST taxonomic rank? A. Family
B. Genus
C. Species
D. Order

Q25.

Which statement about plant taxonomy is CORRECT? A. It is not important in pharmacy
B. It helps in correct identification of crude drugs
C. It deals only with plant names
D. It ignores microscopic characters

🔑 Answer Key (End of File)

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End of Exam